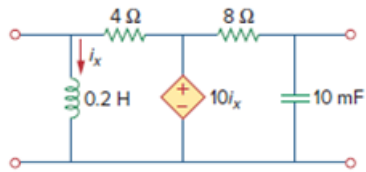


Problem

Obtain the z parameters of the circuit in Fig. 19.54 at $f = 60$ Hz.

Figure 19.54



Step-by-step solution

Step 1 of 6

The impedance parameters are

$$V_1 = Z_{11}I_1 + Z_{12}I_2$$

$$V_2 = Z_{21}I_1 + Z_{22}I_2$$

We use ac analysis at $f = 60$ Hz, because L and C are frequency dependent. By the general equation

$$Z_{11} = \left. \frac{V_1}{I_1} \right|_{I_2=0}$$

$$Z_{21} = \left. \frac{V_2}{I_1} \right|_{I_2=0}$$

If we let $I_1 = 1$ A and open-circuit the output port so that $I_2 = 0$, then we obtain

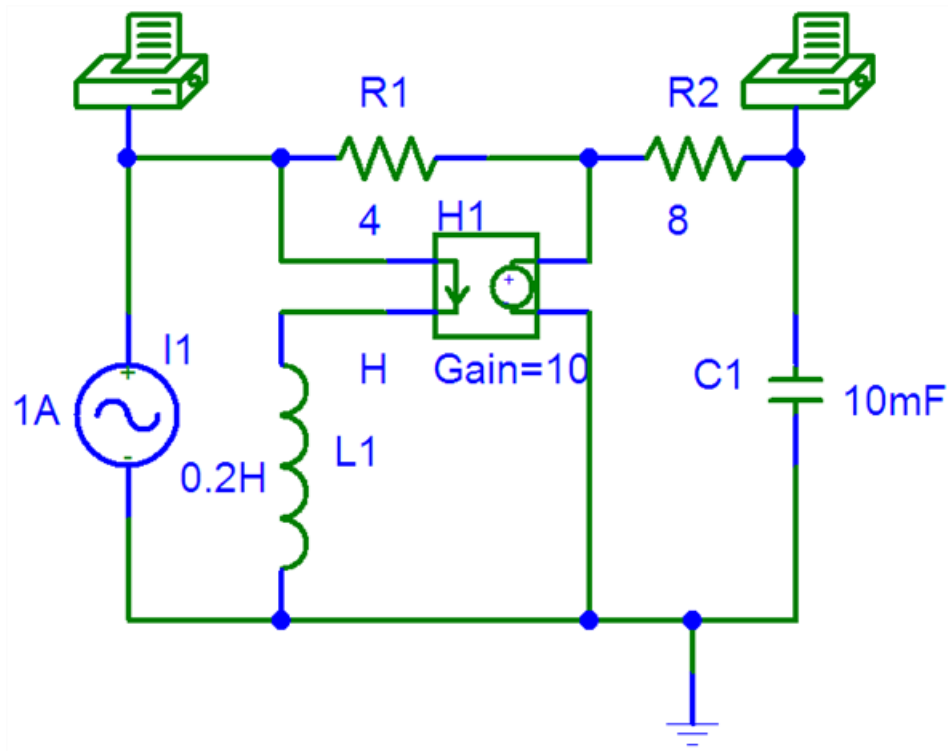
$$Z_{11} = \frac{V_1}{1}$$

$$Z_{21} = \frac{V_2}{1}$$

We insert a 1 A ac current source IAC at the input terminal of the circuit and VPRINT pseudo component to obtain V_1 and V_2 . The attributes of each VPRINT 1 are set as AC=yes,

MAG=yes and PHASE=yes to print the magnitude and phase values of the voltages. Thus, the schematic is

Step 2 of 6



Step 3 of 6

We select analysis/setup/AC sweep and enter 1 as total pts, 60 as start freq and End freq in the AC sweep and noise analysis dialog box. Save the schematic, then we select analysis/simulate to stimulate it. We obtain V_1 and V_2 from the output file.

FREQ VM(\$N_0004) VP(\$N_0004)

6.000E+01 1.753E-02 -2.651E+00

FREQ VM(\$N_0002) VP(\$N_0002)

6.000E+01 3.987E+00 1.755E+02

Thus,

$$Z_{11} = \frac{V_1}{I_1}$$

$$Z_{11} = 3.987 \angle 175.5^\circ \Omega$$

$$Z_{21} = \frac{V_2}{I_1}$$

$$Z_{21} = 0.0175 \angle -2.65^\circ \Omega$$

Step 4 of 6

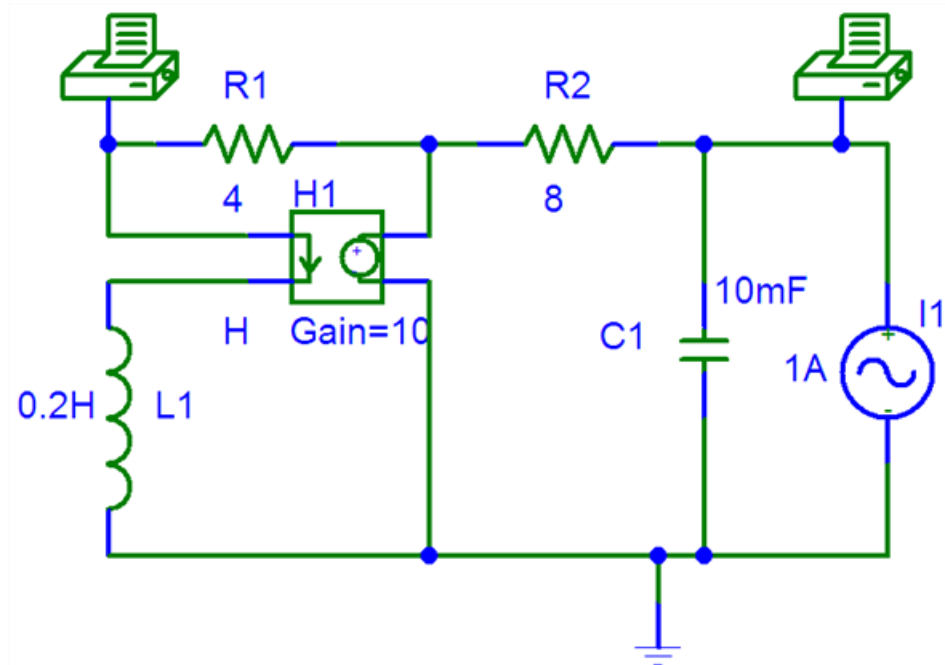
Similarly, from the general equation

$$Z_{12} = \frac{V_1}{I_2} \bigg|_{I_1=0}$$

$$Z_{22} = \frac{V_2}{I_2} \bigg|_{I_1=0}$$

We obtain Z_{12} and Z_{22} by only a difference by the first schematic is that the 1A *ac* current source **IAC** is now at the output terminal. Thus the schematic is

Step 5 of 6



Step 6 of 6

Save the schematic, we select analysis/simulate to simulate it, and we obtain V_1 and V_2 from the output file.

FREQ VM(\$N_0003) VP(\$N_0003)

6.000E+01 2.651E-01 9.190E+01

FREQ VM(\$N_0002) VP(\$N_0002)

6.000E+01 1.000E-30 0.000E+00

Thus

$$Z_{12} = \frac{V_1}{1}$$

$$Z_{12} = 0$$

$$Z_{22} = \frac{V_2}{1}$$

$$Z_{22} = 0.2651 \angle 91.9^\circ \Omega$$

Therefore, the Z parameters are

$Z_{11} = 3.987 \angle 175.5^\circ \Omega$ $Z_{12} = 0$ $Z_{21} = 0.0175 \angle -2.65^\circ \Omega$ $Z_{22} = 0.2651 \angle 91.9^\circ \Omega$
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